

Marvin Gülhan



EDUCATION

University of Bonn <i>Master of Science in Computer Science</i>	Bonn, Germany Oct. 2025 – present
University of Bonn <i>Bachelor of Science in Computer Science (Grade: 1.2)</i>	Bonn, Germany Oct. 2022 – Sep. 2025
Rhein-Sieg-Gymnasium <i>Abitur (Grade: 1.0)</i>	Sankt Augustin, Germany Aug. 2014 – June 2022
German International School of Silicon Valley <i>Semester Abroad</i>	Mountain View (CA), USA Aug. 2019 – Jan. 2020

WORK EXPERIENCE & INTERNSHIPS

Research Assistant <i>Fraunhofer Institute FKIE</i> - Algorithm engineering for online graph problems - Research on RL algorithms for dynamic planning problems	Wachtberg, Germany Nov. 2023 – present
Formula Student Team Member <i>Bonn-Rhein-Sieg University of Applied Sciences (HBRS)</i> - Member of the Formula Student team at HBRS - CFD simulation and aerodynamic design of the rear wing for the 2024 race car	Sankt Augustin, Germany Sep. 2023 – Jun. 2025
Student Internship <i>German Aerospace Center (DLR)</i> - Insights into aerospace engineering and research methods - Use of clustering methods for anomaly detection in fuel combustion of hybrid engines	Cologne, Germany Jan. 2020

PUBLICATIONS & PROJECTS

Reinforcement Learning Amplifies Emergent Misalignment from Harmless Rewards - Co-author of a paper submitted to the 2026 Empirical Methods in Natural Language Processing (EMNLP), Budapest, Hungary - Shows that reinforcement learning (GRPO, preceded by a brief SFT warmup) can broadly misalign a language model not only when rewarding overtly harmful behavior but also when the reward signal looks harmless. Suggests misspecified rewards are a realistic path to misalignment.	2026
Multi-Agent Reinforcement Learning for Complex and Dynamic Graph-Based Planning Problems - Co-author of a peer-reviewed paper accepted at the 2026 International Conference on Military Communications and Information Systems (ICMCIS), Bath, UK - Demonstrated that Graph Reinforcement Learning policies can effectively solve complex, dynamic planning problems such as the Airlift Planning Problem (in-press)	2026
Airlift Challenge: A Competition for Optimizing Cargo Delivery - Co-author of a research paper on optimizing cargo delivery for Airlift Operations - Developed a routing algorithm coordinating heterogeneous aircraft across subgraphs to manage bottlenecks and disruptions in dynamic conditions, achieving 2nd place out of 40 teams	2025

HONORS

Deutschlandstipendium <i>Federal Ministry of Education and Research</i> Merit-based scholarship for outstanding academic performance	Bonn, Germany Oct. 2024 & Oct. 2025
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CORE SKILLS

Programming Languages: Python, C++, Java
Technical Skills: Graph RL, Multi-Agent RL, GNNs, PyTorch, HPC, CFD simulations
Languages: German (native), English (C1), French (B1), Spanish (B1)